

Week 06: Finding useful information

Generative AI | MSc teaching notes and further reading | 2027

Learning target

Compare search strategies while preserving source and access information.

Context and teaching plan

Five fictional college policies must answer student questions.

Contact time	Suggested use
Two hours: concepts	Case introduction, four ideas, worked example and misconception.
Two hours: practice	10 min recall, 20 min demo, 40 min guided practice, 25 min independent change, 15 min review, 10 min exit check.

Explain the key ideas

Chunk

A short passage from a larger source; keep its document ID and version.

Teacher question: Where does chunk appear in the weekly case, and what could go wrong if we omit it?

Keyword search

Matches terms in a query and passage; useful when wording overlaps.

Teacher question: Where does keyword search appear in the weekly case, and what could go wrong if we omit it?

Vector search

Compares numerical representations of text; a production embedding model can capture semantic similarity.

Teacher question: Where does vector search appear in the weekly case, and what could go wrong if we omit it?

Access filter

Only retrieve passages that the current role is allowed to read.

Teacher question: Where does access filter appear in the weekly case, and what could go wrong if we omit it?

Worked example and answer guide

- Input: P2 says library loans may be extended once.
- Change or question: Question uses the phrase renew a book.
- Supported observation: Exact overlap may miss P2; a simple synonym mapping improves this case but can create other misses.

Reasoning: state the given evidence first, then the calculation or rule, then the conclusion. If a value is absent, say what cannot be concluded. Keep any simulated mechanism distinct from a trained model or deployed service.

Teach the week as a story

The library-renewal search: A learner asks “Can I renew my book?” Policy P2 says “library loans may be extended once.” Which passage should search find?

Ask for a quick prediction before each technical explanation. Return to the prediction after the example and ask what evidence changed the answer. The analogies below are memory aids, and each has a stated limit.

Chunk: analogy, mechanism and check

Familiar picture: A long textbook gets labelled index cards for individual sections.

Familiar part	Technical meaning
Book section	passage chunk
Card label	source ID
Edition	version

Where this analogy breaks: Cutting at arbitrary length can separate a condition from its rule.

Actual idea: A short passage from a larger source; keep its document ID and version.

Work it aloud: P2 becomes a short passage with ID P2 and version; keep “once” in the same passage.

Ask before revealing: Why preserve the source ID and version?

Supported response: They let a learner verify the answer in the correct policy edition.

Look for this misconception: A chunk needs only its text.

Keyword search: analogy, mechanism and check

Familiar picture: An index finds pages that contain the exact words you entered.

Familiar part	Technical meaning
Search phrase	query tokens
Word matches	overlap score

Top cards	ranked passages
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Where this analogy breaks: Synonyms can make a relevant passage score poorly.

Actual idea: Matches terms in a query and passage; useful when wording overlaps.

Work it aloud: “renew” does not exactly match P2’s “extend”; a keyword-only search may miss it.

Ask before revealing: What explains this miss?

Supported response: Different words express a similar intent.

Look for this misconception: No exact match proves no relevant policy exists.

Vector search: analogy, mechanism and check

Familiar picture: A map places related meanings near one another.

Familiar part	Technical meaning
Question	numeric embedding
Policy	numeric embedding
Distance	similarity

Where this analogy breaks: Similar vectors can retrieve the wrong passage; check text and permissions.

Actual idea: Compares numerical representations of text; a production embedding model can capture semantic similarity.

Work it aloud: Toy vectors $q=[1,0]$, $P2=[1,1]$, $P3=[0,1]$: cosine scores about .707 and 0.

Ask before revealing: Which passage ranks nearer to q in this toy calculation?

Supported response: P2, with the larger cosine similarity.

Look for this misconception: An embedding is an authoritative answer.

Access filter: analogy, mechanism and check

Familiar picture: A library card allows entry to some shelves but not staff cabinets.

Familiar part	Technical meaning
Student role	access identity
Open shelf	allowed policy
Cabinet	restricted source

Where this analogy breaks: Filtering after generation may leak details already inserted into context.

Actual idea: Only retrieve passages that the current role is allowed to read.

Work it aloud: Filter staff-only P5 before ranking; never supply it to the student answer stage.

Ask before revealing: When must the access filter apply?

Supported response: Before retrieval results are passed into an answer.

Look for this misconception: A relevant staff policy may be shown if cited.

New situation for discussion

A staff-only policy has a perfect keyword match. Should the student see it?

Memory cue: Split with source IDs, compare retrieval methods, and filter permissions first.

Guided laboratory

Compare token overlap and the synonym proxy on a fixed corpus.

1. Create a five-policy corpus with access labels
2. Search ten questions with token overlap
3. Use the supplied synonym expansion as a meaning-aware proxy
4. Measure whether the known policy is in the top two

Individual change: Remove a synonym or lower top-k from 2 to 1 and compare recall. Expected evidence: Corpus manifest and retrieval misses with evidence IDs.

Checks and common error

Measure: Recall@2 on ten labelled questions. Record the input, setting, source version, result and the relevant numerator and denominator.

Plausible incorrect claim: The synonym mapping is a production embedding model.

Correction: It is an inspectable proxy for class discussion. A vector index would require a tested embedding model.

If learners disagree, reproduce the same case and compare the underlying source or expected answer. Record unresolved cases for instructor review.

Further discussion

- How could the conclusion change if the source or input changes?
- Which part of the result is observed and which part is an assumption?
- What simple test would detect the failure described above?

References and optional extension

Core readings are official vendor documentation or original research. Read the relevant section; an account or paid API is unnecessary for this notebook. Page details may change before 2027.

Microsoft, Retrieval augmented generation.

<https://learn.microsoft.com/en-us/azure/foundry/concepts/retrieval-augmented-generation>

Lewis et al., Retrieval-Augmented Generation. <https://arxiv.org/abs/2005.11401>

Optional extension: replace one classroom proxy with a locally approved model or service, keep the original baseline, repeat the same checks, and state the additional privacy and cost assumptions.

Curriculum scope: the supplied 12-week teaching sequence, recorded in docs/SYLLABUS_ALIGNMENT.md. The original module descriptor is retained in sources/originals/07-Descriptor-Generative-AI-new-.docx. Supporting project notes: Deep-research.txt, RAG.txt, AI-Agents.txt, MCP-&-Vector-Database.txt, Multi-Agent-AI-System.txt, Eval-&-Fine-tune-&-Loop-Engineering.txt. These are planning sources, not empirical performance evidence.

Student revision and exam preparation

Outcomes: LO4, LO5 | 2 hours concepts + 2 hours guided practice

The idea in plain language

Split sources into passages while preserving document IDs, versions and roles. Keyword search depends on wording. Embeddings place inputs in a numerical representation where similarity can help retrieval. Filter permissions before selecting evidence. The class synonym search is a transparent proxy, not a learned embedding model.

A worked example

The query renew my book and policy library loans may be extended use different words. Keyword overlap may miss the source. A meaning-aware representation can help, but its ranking must still be evaluated. If P5 is staff-only, a student query must not retrieve it.

Practice before reading the answers

1. Why keep source IDs and versions with chunks? (4 suggested practice marks)
2. One relevant passage exists and appears in the top two. What is recall@2? (4 suggested practice marks)
3. Can a high similarity score override a role restriction? (4 suggested practice marks)

Draw the main diagram from memory, explain each arrow, then check it against the slides. Answer the three questions without notes before reading the model answers.

Model answers and marking guidance

1. To trace an answer to the right document and identify stale evidence.
2. $1/1 = 1$ for that query. Average across queries with relevant evidence; handle no-answer cases separately.
3. No. Similarity is about relevance; permission determines allowed access.

For each four-mark practice answer: two marks for the correct idea, one for a relevant example or calculation, one for an appropriate explanation or limitation. These suggested practice marks do not change the module assessment scheme.

Official reading and video support

M1 Microsoft Learn: Introduction to retrieval-augmented generation concepts

<https://learn.microsoft.com/en-us/training/modules/rag-fundamentals/>

Study focus: Preparation, retrieval, generation and evaluation.

M2 Microsoft Learn: RAG and Vector Databases: Generative AI for Beginners

<https://learn.microsoft.com/en-us/shows/generative-ai-for-beginners/retrieval-augmented-generation-rag-and-vector-databases-generative-ai-for-beginners>

Study focus: Optional RAG overview; hosted on Microsoft Learn.

Selected source concepts inform the lesson. Examples, figures, questions and solutions are original classroom material; no exam questions are claimed to be official.